

A Crash Course in Integration in $\mathbb{R}^2$ and $\mathbb{R}^3$

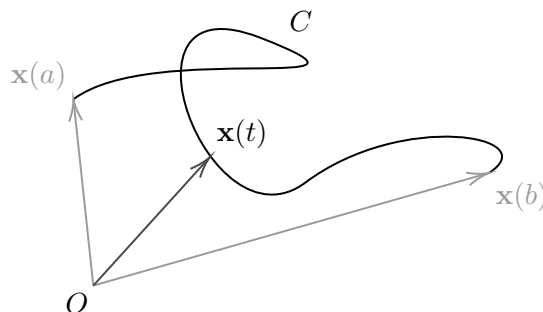
CircleThm#7102

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This is a no-frills description of how to integrate over curves, areas, volumes and surfaces in $\mathbb{R}^2$ and $\mathbb{R}^3$. Very little will be actually motivated, but it should (hopefully) still be helpful.

1 The Line Integral

Suppose you have a regular curve C , parametrised by some function $\mathbf{x}(t)$, where $a \leq t \leq b$, and you want to integrate some other function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ over the curve.



Then we have, integrating from $\mathbf{x}(a)$ to $\mathbf{x}(b)$,

$$\int_C f(\mathbf{x}) ds = \int_a^b f(\mathbf{x}(t)) \left| \frac{d\mathbf{x}(t)}{dt} \right| dt.$$

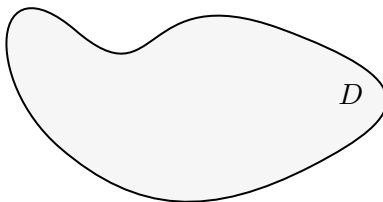
If instead we wanted to integrate over some vector field (by say taking the dot product of that vector field with the unit tangent), then since $d\mathbf{x}(t)/dt$ is tangent to the curve, we would have

$$\int_C F(\mathbf{x}) \cdot d\mathbf{x} = \int_a^b \mathbf{F}(\mathbf{x}(t)) \cdot \frac{d\mathbf{x}}{dt} dt,$$

where again we integrate from $\mathbf{x}(a)$ to $\mathbf{x}(b)$.

2 Area and Volume Integrals

Suppose you have some closed region $D \subset \mathbb{R}^2$, and you wanted to integrate some function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ over the region.



Then integrate over say the x coordinate first (where the bounds of the inner integral will depend on y), we have

$$\int \int_D f(\mathbf{x}) dA = \int \int_{(x,y) \in D} f(x,y) dx dy$$

Of course we could have integrated over the y coordinate first, and would probably give the same result.

We can also do a change of coordinate. Suppose we have $x = x(u, v)$ and $y = y(u, v)$ being smooth bijections (with smooth inverse) mapping D' in the (u, v) -plane to the region D in the (x, y) -plane. Then we have

$$\int \int_{(x,y) \in D} f(x,y) dx dy = \int \int_{(u,v) \in D'} f(x(u,v), y(u,v)) |J| du dv,$$

where

$$|J| = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \left| \det \left(\frac{\partial \mathbf{x}}{\partial u} \mid \frac{\partial \mathbf{x}}{\partial v} \right) \right| = \left| \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} \right|$$

is the *Jacobian*.

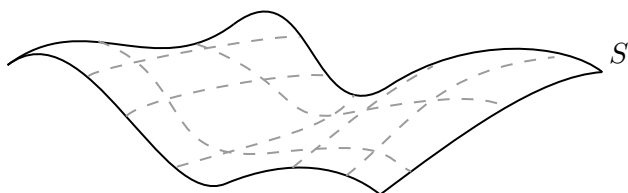
Basically the exact same thing is true for volume integrals – I won't bore you with the details. Just use three variables and change the Jacobian accordingly.

3 Surface Integrals

Suppose you have a surface S , parametrised with

$$S = \{\mathbf{x}(u, v) \mid (u, v) \in D\},$$

for some region D in the (u, v) -plane. Suppose also that you wanted to integrate some function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ over the surface.



Then we would have

$$\int \int_S f(\mathbf{x}) dS = \int \int_D f(\mathbf{x}(u, v)) \left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right| du dv.$$

If instead we wanted to integrate over some vector field $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by taking the dot product of the vector field with the unit normal (a flux integral), then since $\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v}$ is normal to the curve, we would have

$$\int \int_S \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S} = \int \int_D \mathbf{F}(\mathbf{x}(u, v)) \cdot \left(\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right) du dv.$$

It's worth noting here that by doing this, we implicitly chose a direction for the normal (there's 2!). If we wanted the other direction, we can just negate the integral.

4 Integral Theorems

Having read that, you are clearly now an expert integrator, keen to speed up your multidimensional calculations. Luckily we have a few results at our disposal which we will state (naturally) without proof. Throughout, we will use the notation ∂A to denote the boundary of A .

Theorem 4.1 (Conservative Vector Fields). *Suppose the vector field $\mathbf{F}$ can be written as $\mathbf{F} = \nabla f$ for some scalar function f . Then if C is any curve from $\mathbf{x}(a)$ to $\mathbf{x}(b)$, we have*

$$\int_C \mathbf{F} \cdot d\mathbf{x} = f(\mathbf{x}(b)) - f(\mathbf{x}(a)).$$

Theorem 4.2 (Green's Theorem). *$P = P(x, y)$ and $Q = Q(x, y)$ are continuously differentiable functions on A and ∂A is made from a collection of piecewise smooth curves, then*

$$\int_{\partial A} P \, dx + Q \, dy = \iint_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy.$$

The orientation of the boundary ∂A is such that A lies to your left as you traverse it.

Theorem 4.3 (Stokes' Theorem). *If $\mathbf{F} = \mathbf{F}(\mathbf{x})$ is a continuously differentiable vector field and S is an orientable, piecewise regular surface with piecewise regular boundary ∂S then*

$$\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{x}.$$

Theorem 4.4 (Divergence Theorem in $\mathbb{R}^3$). *If $\mathbf{F} = \mathbf{F}(\mathbf{x})$ be a continuously differentiable vector field and V is a volume with a piecewise regular boundary ∂V then*

$$\int_V \nabla \cdot \mathbf{F} \, dV = \int_{\partial V} \mathbf{F} \cdot d\mathbf{S}$$

where the normal to ∂V points outwards from V .

Theorem 4.5 (Divergence Theorem in $\mathbb{R}^2$). *Let $\mathbf{F} = \mathbf{F}(\mathbf{x})$ be a continuously differentiable vector field and $D \subset \mathbb{R}^2$ region with piecewise smooth boundary ∂D then*

$$\int_D \nabla \cdot \mathbf{F} \, dA = \oint_{\partial D} \mathbf{F} \cdot \mathbf{n} \, ds$$

where the unit normal $\mathbf{n}$ to ∂D points outwards from D .